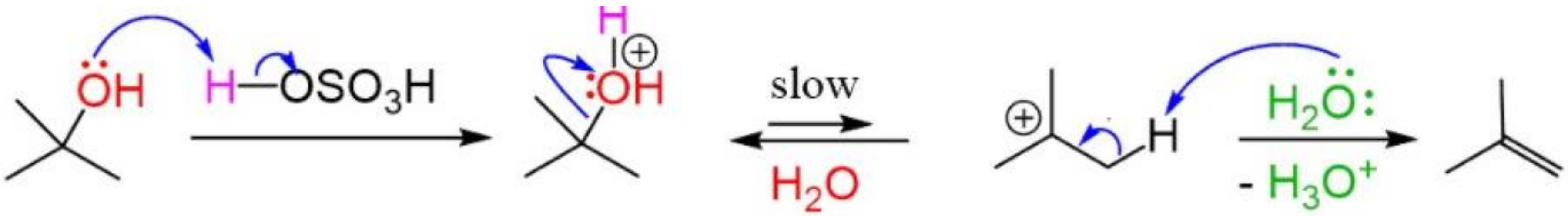
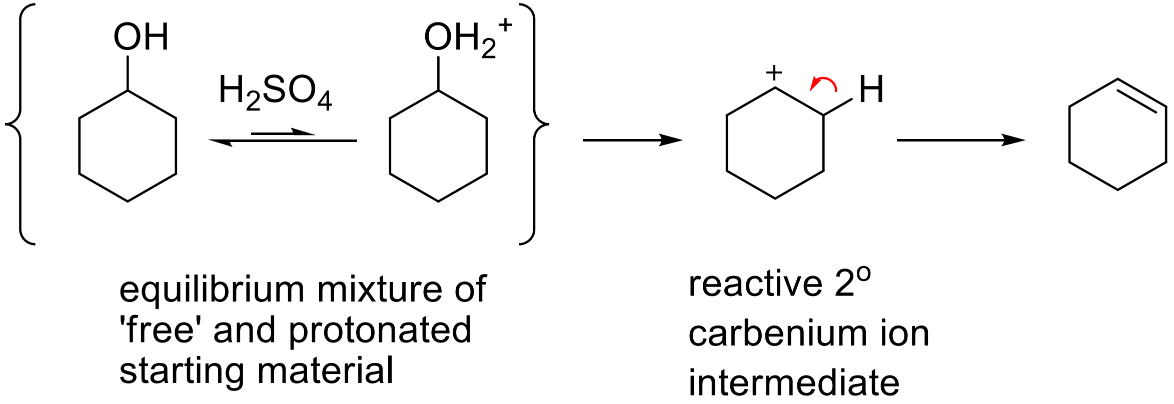
Why is E2 not easy for alcohols?

Why is E2 not easy for alcohols?

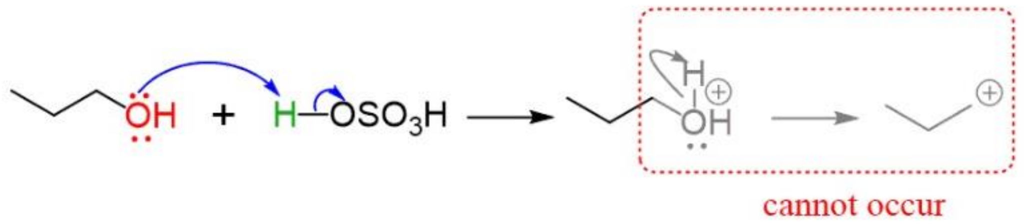
Secondary and tertiary alcohols will form more stable carbocations and will undergo E1 elimination rather than E2 elimination. For Example:



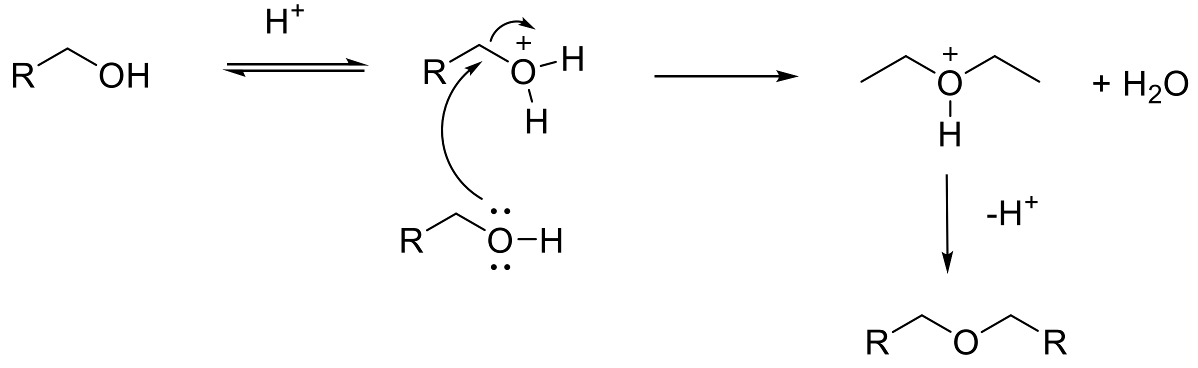


Why is E2 not easy for alcohols?

In the case of primary alcohols E1 is not likely to happen, primary carbocations are not very stable. Elimination reaction will happen via E2 mechanism. SN2 reaction of primary alcohols with themselves can compete.



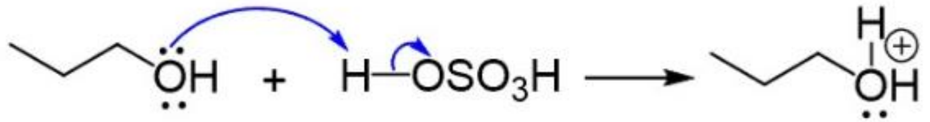
E1 ELIMINATION CANNOT OCCUR, LOW STABILITY OF 1^{O} CARBOCATION



SN2 REACTION WITH ITSEF COMPETES WITH E2

Dehydration of Primary Alcohols by E2 mechanism

Step 1. *Protonation of the hydroxyl group*



Step 2. *Dehydration of alcohol by E2 mechanism*

WEAK BASE AVAILABLE
REACTION WITH HIGH ENERGY BARRIER WILL REQUIRE HIGH TEMPERATURE

Draw the structure of 2-butene. HCl was induced to react with 2-butene. Write the mechanism of the reaction that proceeds of HCl with 2-butene. Give the name of the chlorinated product that is formed.

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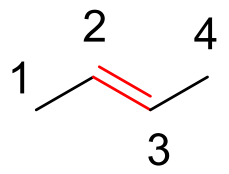
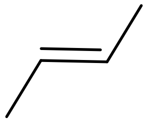
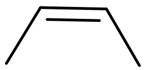


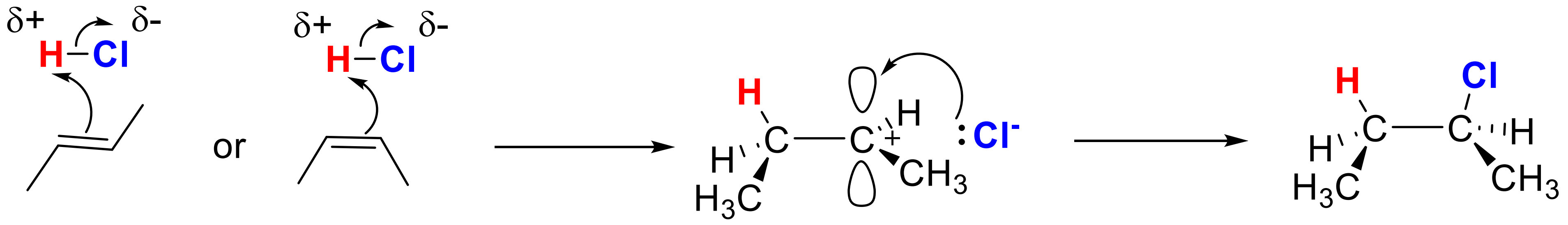
2-butene, this alkene can be *cis* or *trans*. The mos complete answer is:



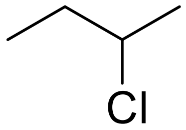
trans-2-butene



cis-2-butene

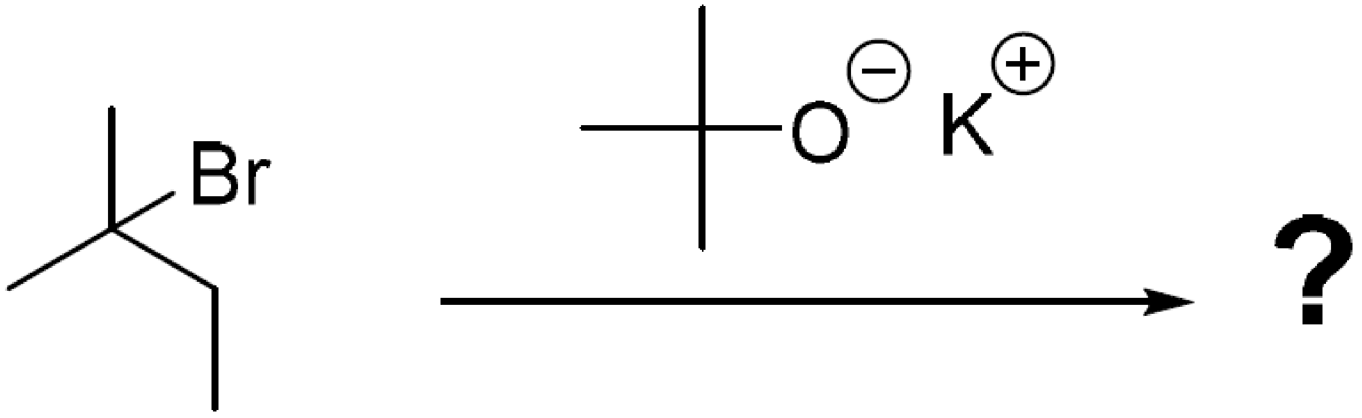


both isomers
lead to the same carbocation

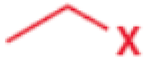
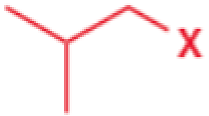
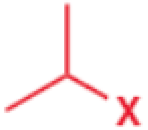
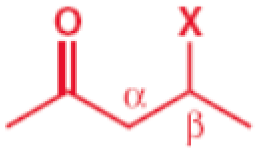


2-chloro-
butane

For the reaction below, draw the mechanism, give the structure of all the organic products that are formed indicating which one, if any, is formed as a major product stating why.

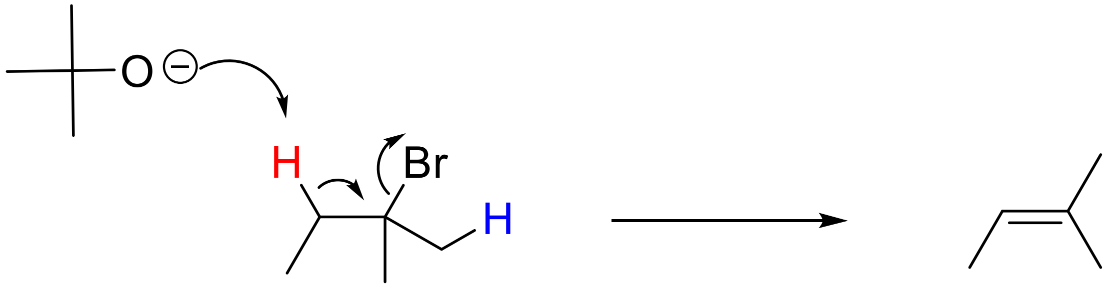


For the reaction below, draw the mechanism, give the structure of all the organic products that are formed indicating which one, if any, is formed as a major product stating why.

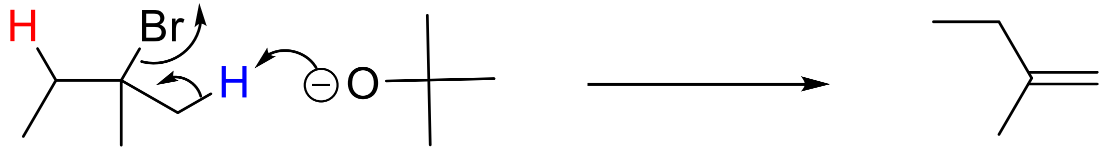
		Poor nucleophile (e.g. H_2O, ROH)	Weakly basic nucleophile (e.g. I^-, RS^-)	Strongly basic, unhindered nucleophile (e.g. RO^-)	Strongly basic, hindered nucleophile (e.g. DBU, *t*-BuO^-)
methyl	$H_3C—X$	no reaction	S_N2	S_N2	S_N2
primary (unhindered)		no reaction	S_N2	S_N2	E2
primary (hindered)		no reaction	S_N2	E2	E2
secondary		S_N1, E1 (slow)	S_N2	E2	E2
tertiary		E1 or S_N1	S_N1, E1	E2	E2
β to anion-stabilizing group		E1cB	E1cB	E1cB	E1cB

For the reaction below, draw the mechanism, give the structure of all the organic products that are formed indicating which one, if any, is formed as a major product stating why.

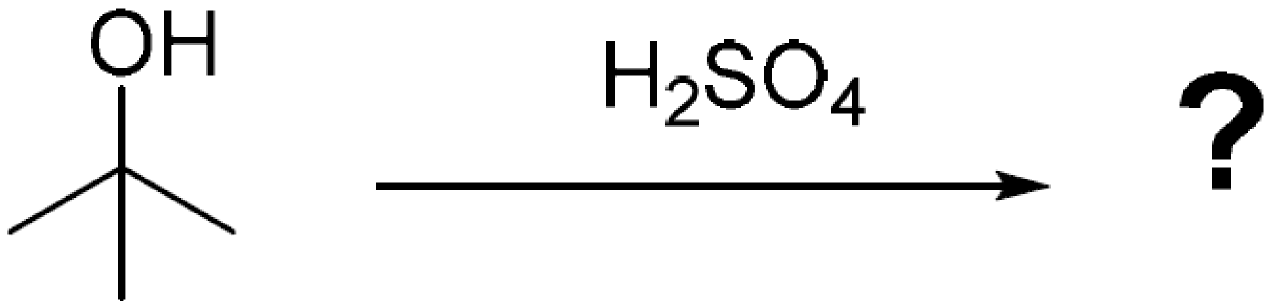
E2 Elimination



Most substituted alkene preferred because is the most stable

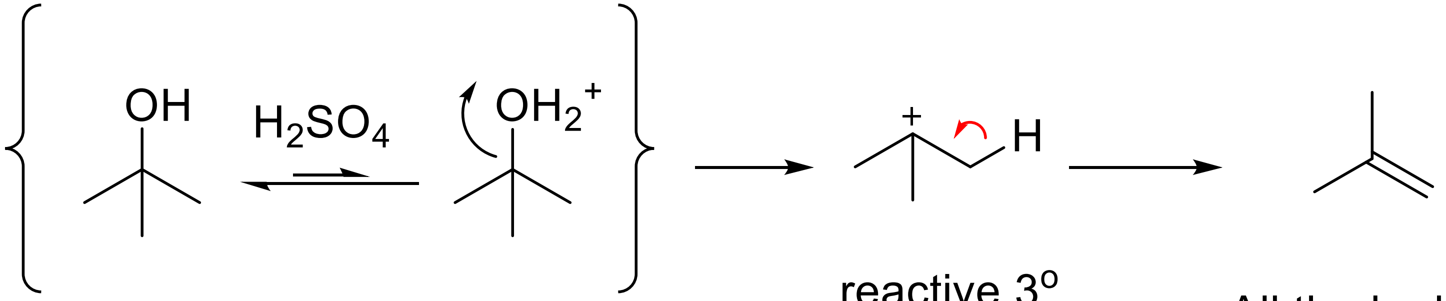


For the reaction below, draw the mechanism, give the structure of all the organic products that are formed indicating which one, if any, is formed as a major product stating why.



For the reaction below, draw the mechanism, give the structure of all the organic products that are formed indicating which one, if any, is formed as a major product stating why.

An alcohol in the presence of an acid with a non-nucleophilic conjugate base is going to undergo Elimination reaction (see notes)
The E2 mecanism is very rare. The elimination will proceed most likely via an E1 mechanism

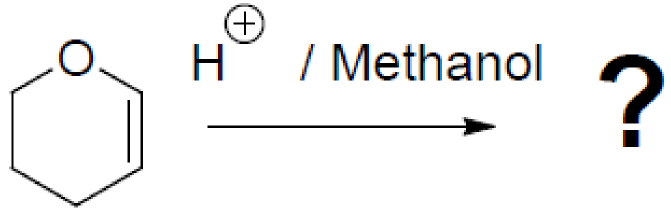


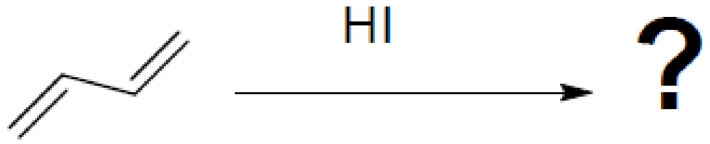
equilibrium mixture of 'free' and protonated starting material

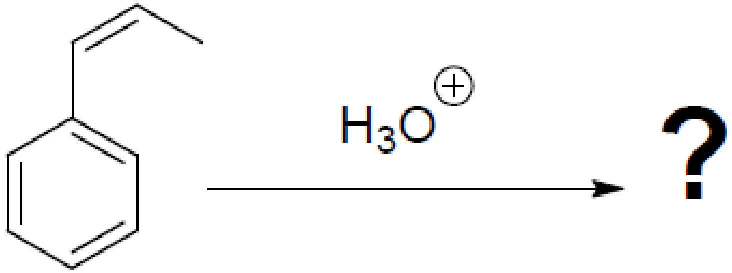
reactive 3^o carbocation intermediate

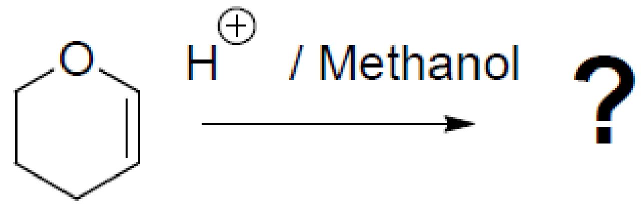
All the hydrogen positions are equivalent and lead to the same product

Propose all possible products and mechanisms for the following reactions:

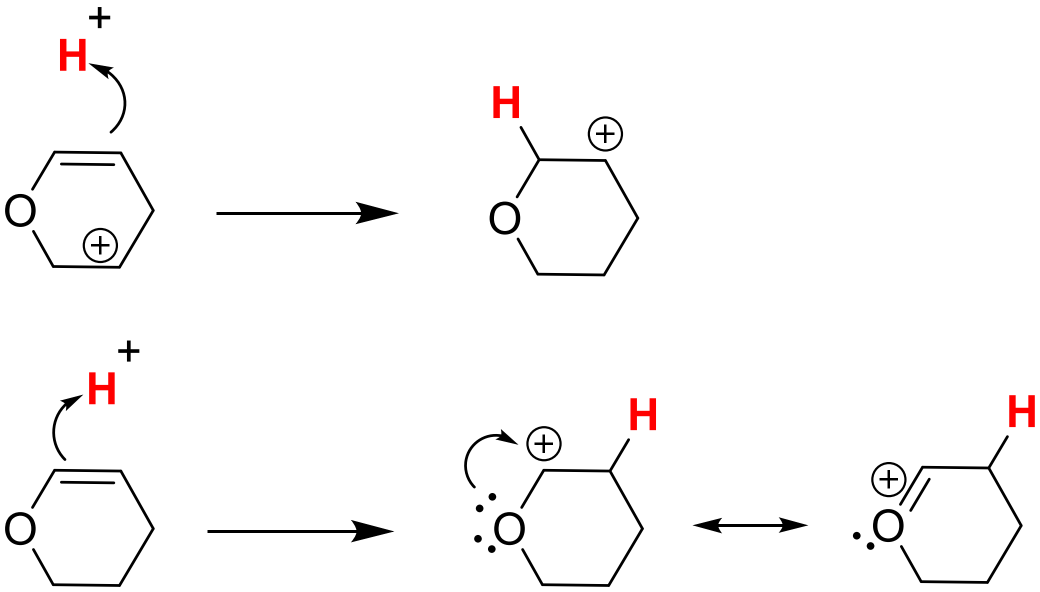






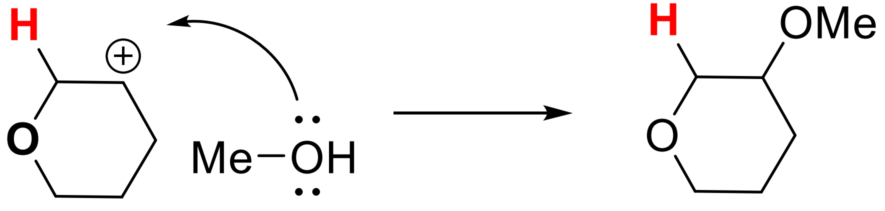


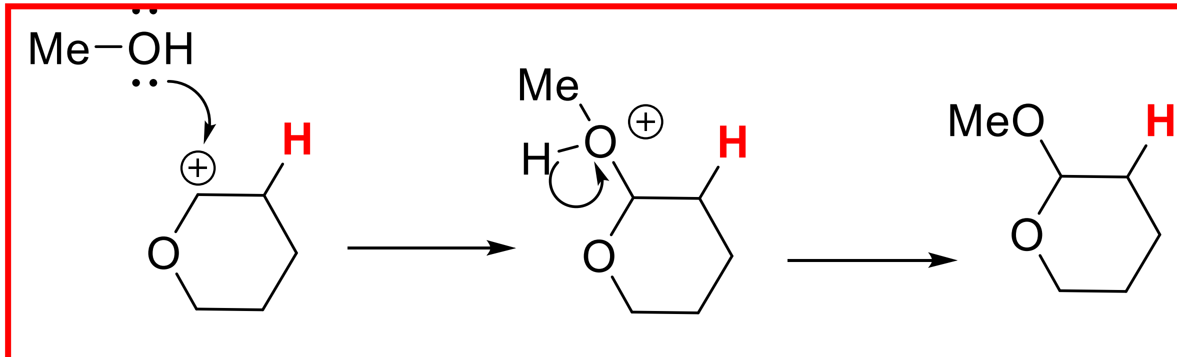
1st step formation of carbocation



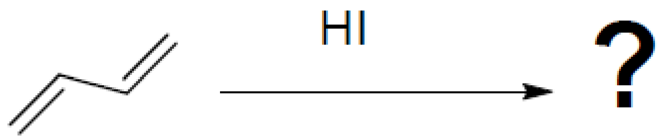
more stable carbocation

2nd step addition of methanol



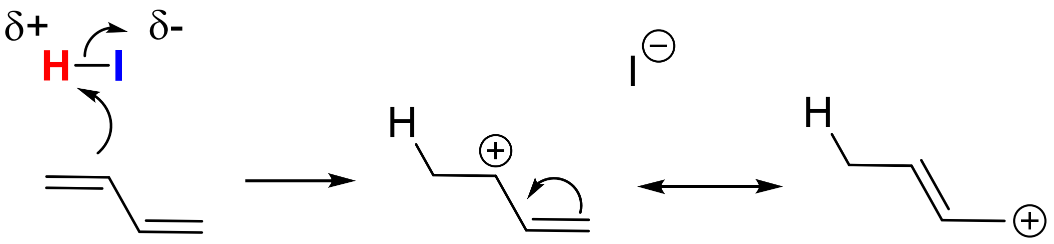


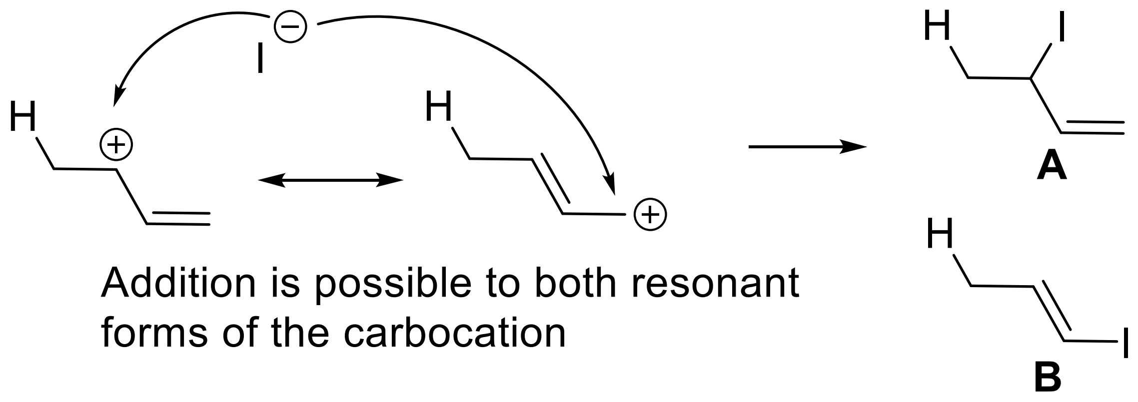
MAJOR PRODUCT



1st step formation of carbocation

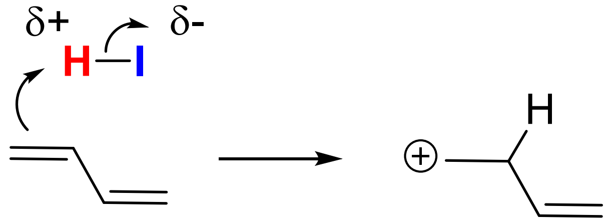
2nd step addition of iodide



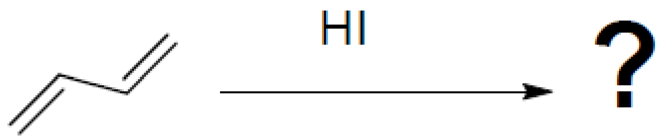


more stable carbocation

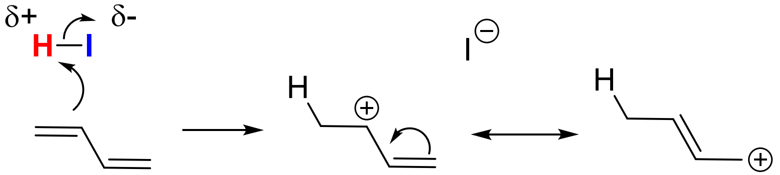
Addition is possible to both resonant forms of the carbocation



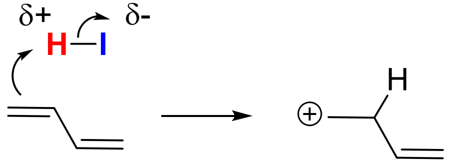
NOT LIKELY TO HAPPEN



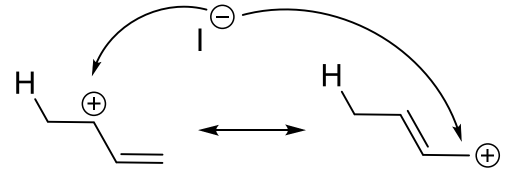
1st step formation of carbocation



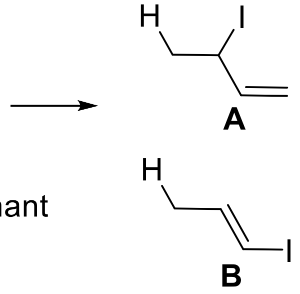
more stable carbocation



2nd step addition of iodide

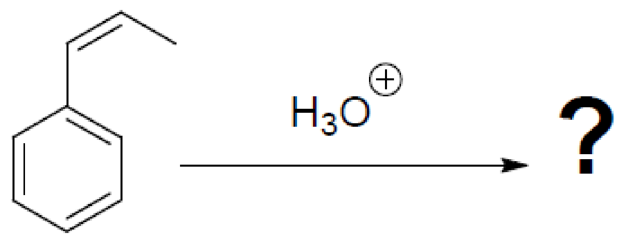


Addition is possible to both resonant forms of the carbocation



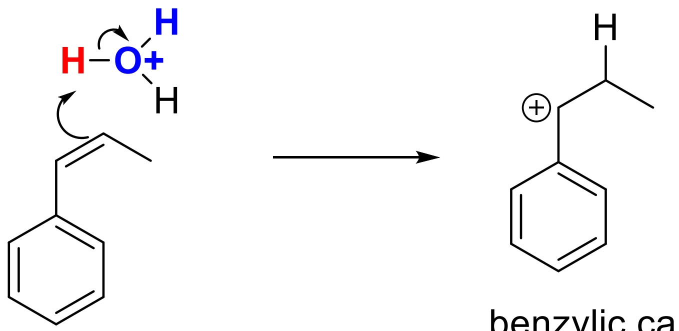
fast, more stable carbocation, lower activation energy (Kinetic control)

slow, less stable carbocation, higher activation energy more stable product, more substituted alkene (Thermodinamic control)

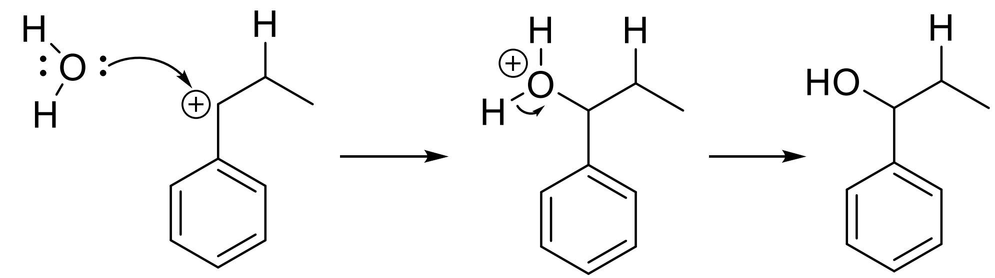


1st step formation of carbocation

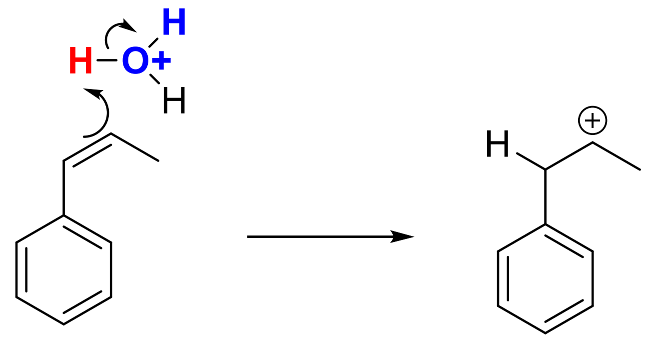
2nd step addition of water



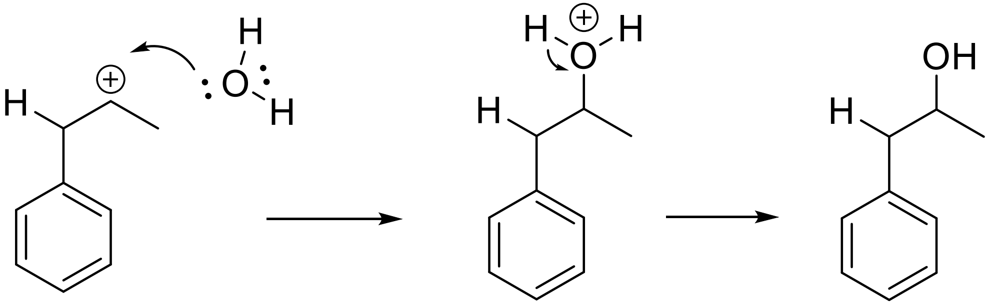
benzylic carbocation is more stable



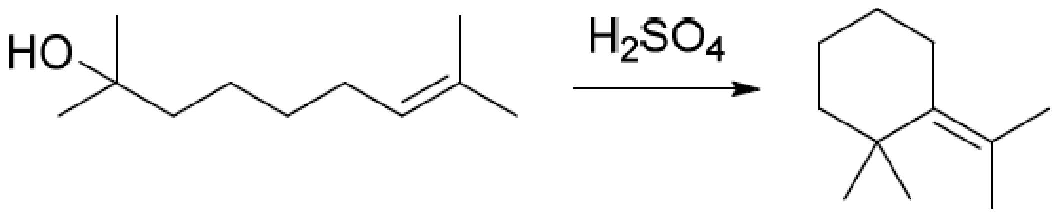
MAJOR PRODUCT



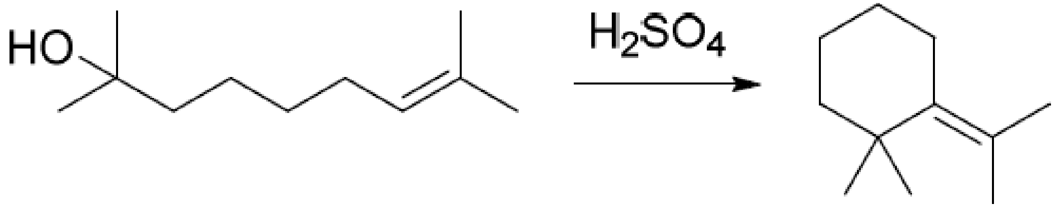
2^o carbocation is less stable



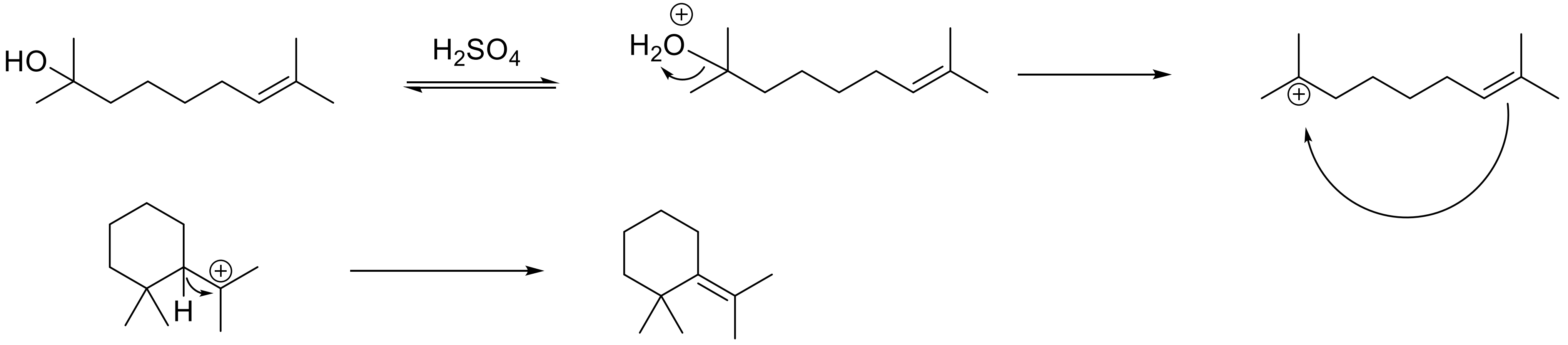
Draw a mechanism for the following reaction:



Draw a mechanism for the following reaction:

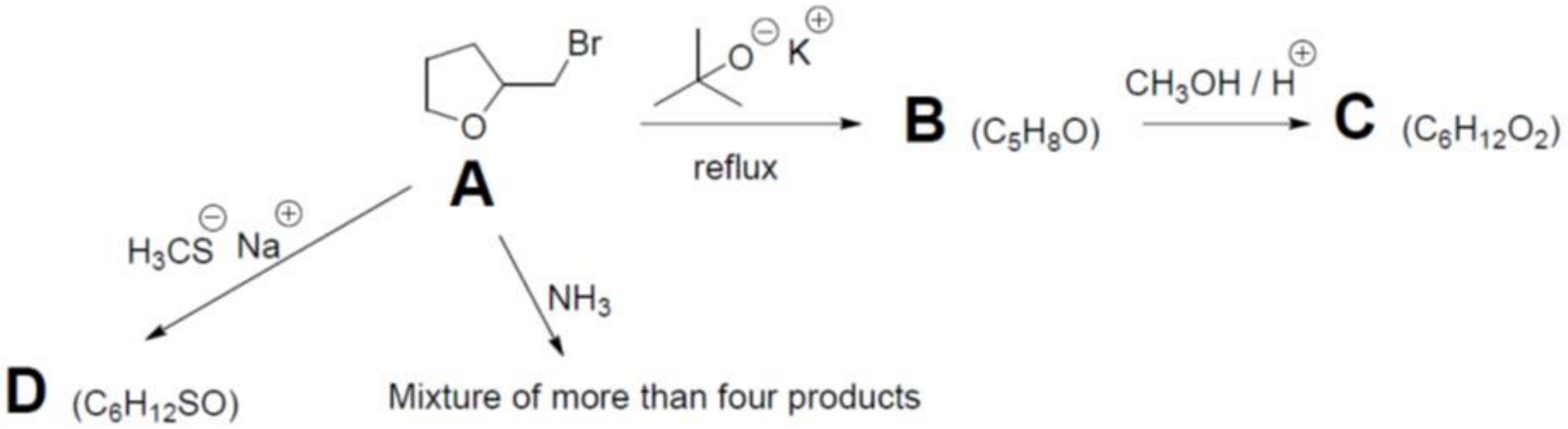


formation of carbocation an intramolecular addition
an alkene can act as a nucleophile

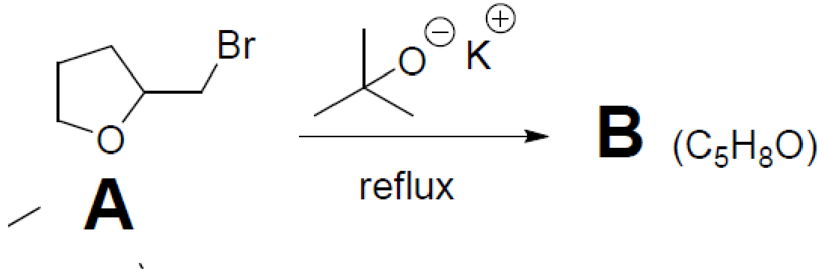


new carbocation ready
to undergo E1 reaction.

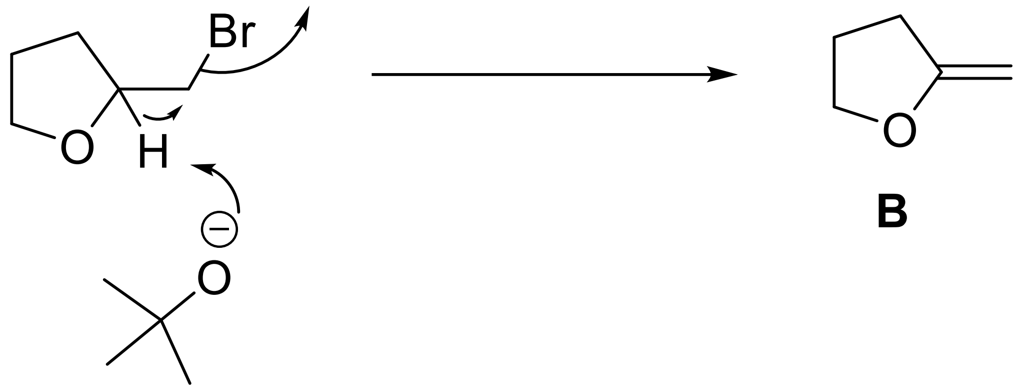
Consider the following reactions:



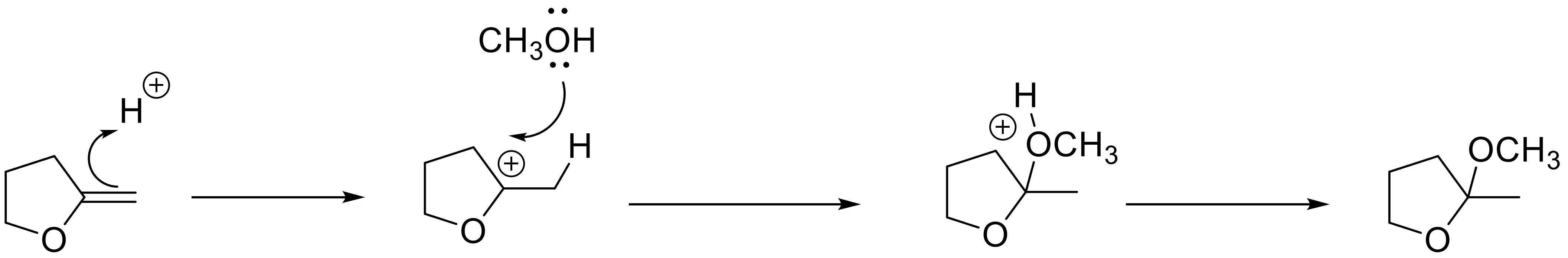
a) Draw the structure of B, C and D.
b) Draw the mechanism of the reaction from A to B
c) Draw the mechanism of the reaction from B to C, make sure you include relevant resonance as part of your
d) mechanism
Draw the mechanism of the reaction from A to D
e) Explain using a mechanism, why the reaction between ammonia and A leads to a mixture of products.



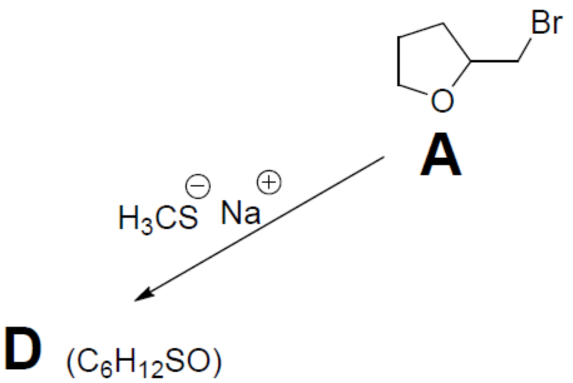
hindered primary carbon in the presence of hindered strong base leads to E2 elimination (see notes)



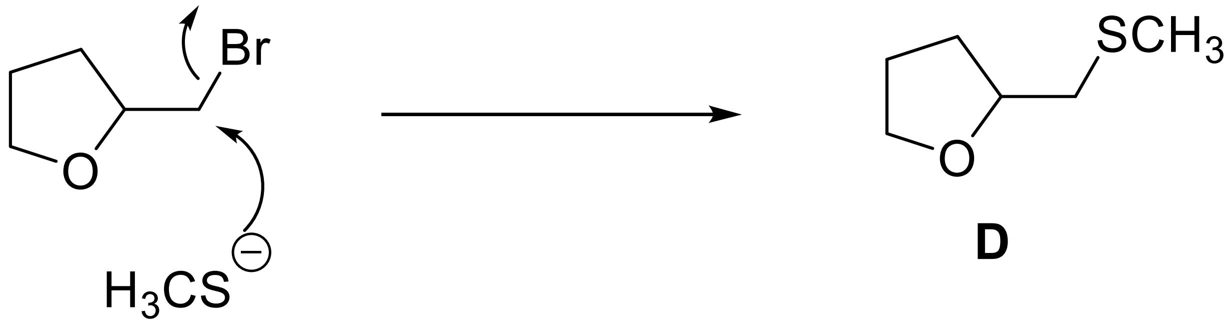
$$\mathbf{B}\ (C_5H_8O) \xrightarrow{CH_3OH\ /\ H^{\oplus}} \mathbf{C}\ (C_6H_{12}O_2)$$

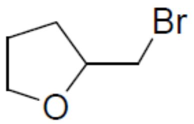


B | addition of H to form the most stablke carbocation | Addition of MeOH | Deprotonation | **C**

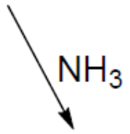


A is a primary hindered alkyl bromide in the presence of a weakly basic nucleophile. These conditions lead to SN2 reaction (See notes)





A



Mixture of more than four products

A is a primary hindered alkyl bromide in the presence of a weakly basic nucleophile NH_3. These conditions lead initially to SN2 reaction.